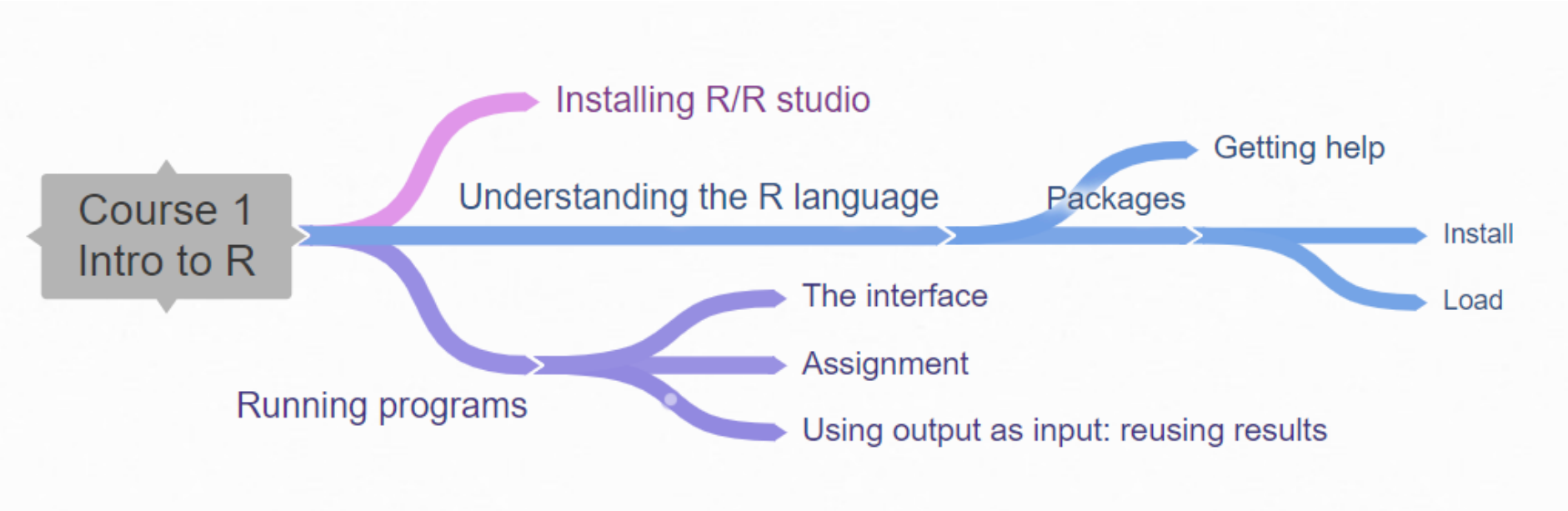
Course 2:
Creating a dataset in R

Review


Course 1
Intro to R
Installing R/R studio
Understanding the R language
Getting help
Packages
Install
Load
Running programs
The interface
Assignment
Using output as input: reusing results

1 Understanding datasets

2 Data structures

3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

6 Summary

A dataset

- Rows: observations
- Columns: variables

PatientID	AdmDate	Age	Diabetes	Status
1	11/15/2015	45	Type1	Poor
2	10/05/2015	67	Type2	Improved
3	12/11/2015	43	Type1	Excellent
4	12/01/2015	56	Type2	Improved

1 Understanding datasets

2 Data structures

3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

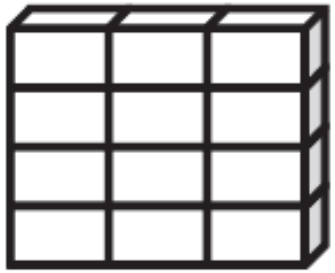
6 Summary

Data structures

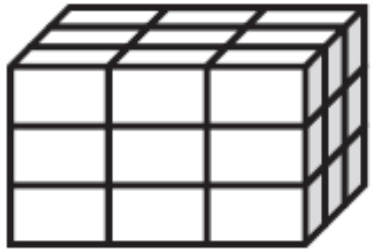
(a) Vector



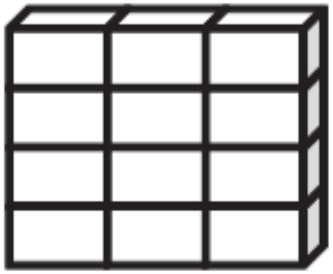
(b) Matrix



(c) Array



(d) Data frame



Columns can be different modes

(e) List

- Vectors
- Arrays
- Data frames
- Lists

Data structures

Some definitions

- Object: anything that can be assigned to a variable, including constants, data structures, functions, and even graphs.
- Data frame: a structure in R that holds data and is similar to the datasets found in SAS, SPSS, and Stata.
- Factors: nominal or ordinal variables.They're stored and treated specially in R.

Important points

1. How to generate a specific data structure in R?
2. How to identify a particular element in a specific data structure?

Data structure

Vector

- One-dimensional arrays that can hold numeric data, character data, or logical data.
- The combine function c( ) is used to form the vector.

```
# a numeric vector
a <- c(1, 2, 8)
# a character vector
b <- c("Poor", "Improved", "Excellent")
# a logical vector
d <- c(TRUE, FALSE, TRUE)
```

Note

Data in a vector must be only one type or mode (numeric, character, or logical). You can't mix modes in the same vector.

Vector

```
a <- c(1, 2, 8)
b <- c("Poor", "Improved", "Excellent")
d <- c(TRUE, FALSE, TRUE)
```

Refer to elements of a vector: [ ]

```
    #8
    a[3]
    #"Improved"
    b[2]
    #TRUE FALSE
    d[1:2]
```

Note: The colon : generates a sequence of numbers. For example, 1:3 is equivalent to 1,2,3

Data structure

Matrices

- A matrix is a two-dimensional array in which each element has the same mode(numeric, character, or logical).
- Matrices are created with the matrix( ) function.
- Matrices are two-dimensional and, like vectors, can contain only one data type.

```
1     mymatrix <- matrix(vector, nrow, ncol, dimnames)
```

- vector contains the elements for the matrix
- nrow and ncol specify the row and column dimensions

Matrices

- Create a 5×4 matrix

```
> mymatrix <- matrix(1:20, nrow=5)
> mymatrix
     [,1] [,2] [,3] [,4]
[1,]    1    6   11   16
[2,]    2    7   12   17
[3,]    3    8   13   18
[4,]    4    9   14   19
[5,]    5   10   15   20
```

- Fill a 2×2 matrix by rows

```
> mymatrix <- matrix(c(1, 3, 5, 7), nrow = 2, byrow = TRUE,
dimnames = list(c("R1", "R2"), c("C1", "C2")))
> mymatrix
   C1 C2
R1  1  3
R2  5  7
```

Matrices

Identification of elements of a matrix

- X[i,]: the ith row of matrix X
- X[, j]: the jth column of matrix X
- X[i, j] refers to the ij^{th} element

```
> mymatrix <- matrix(1 : 20, nrow = 5)
> mymatrix
     [,1] [,2] [,3] [,4]
[1,]    1    6   11   16
[2,]    2    7   12   17
[3,]    3    8   13   18
[4,]    4    9   14   19
[5,]    5   10   15   20
> mymatrix[1, ]
[1]  1  6 11 16
> mymatrix[, 2]
[1]  6  7  8  9 10
> mymatrix[3, 4]
[1] 18
```

What is mymatrix[2, 2:4] or mymatrix[2, c(2:4)]?

Data structures

Data frames

- More general than a matrix in that different columns can contain different modes of data (numeric, character, and so on)
- Data frames are the most common data structure you'll deal with in R

```
1     #col1, col2 are column vectors of any type
2    mydata <- data.frame(col1, col2, ...)
```

Data frames

- Example of creating a data frame:

```
>
> patientID <- c(1:4)
> age <- c(23, 34, 53, 65)
> status <- c("Improved", "Poor", "Excellent", "Improved")
> df_patient <- data.frame(patientID, age, status)
> df_patient
  patientID age    status
1         1  23  Improved
2         2  34      Poor
3         3  53 Excellent
4         4  65  Improved
```

  - Each column has only one mode
  - The data frame is formed by different modes

Data frames

Identify elements of a data frame

1. Use the subscript notation we used before (for example, with matrices): X[i,j]

```
> df_patient[1:2, ]
  patientID age   status
1         1  23 Improved
2         2  34     Poor
> df_patient[ , 1:2]
  patientID age
1         1  23
2         2  34
3         3  53
4         4  65
> df_patient[1, 2]
[1] 23
```

Data frames

Identify elements of a data frame

2. Specify column names

```
> df_patient[, c("age", "status")]
  age    status
1  23  Improved
2  34      Poor
3  53 Excellent
4  65  Improved
> df_patient[ c("age", "status")]
  age    status
1  23  Improved
2  34      Poor
3  53 Excellent
4  65  Improved
> df_patient$age
[1] 23 34 53 65
```

Data frames

The notation $

It's used to indicate a particular variable from a given data frame.

```
> mean(df_patient$age)
[1] 43.75
```

Add a column to a dataset

```
#add a gender variable
df_patient$gender <- c(1, 2, 1, 2)
```

1 Understanding datasets

2 Data structures

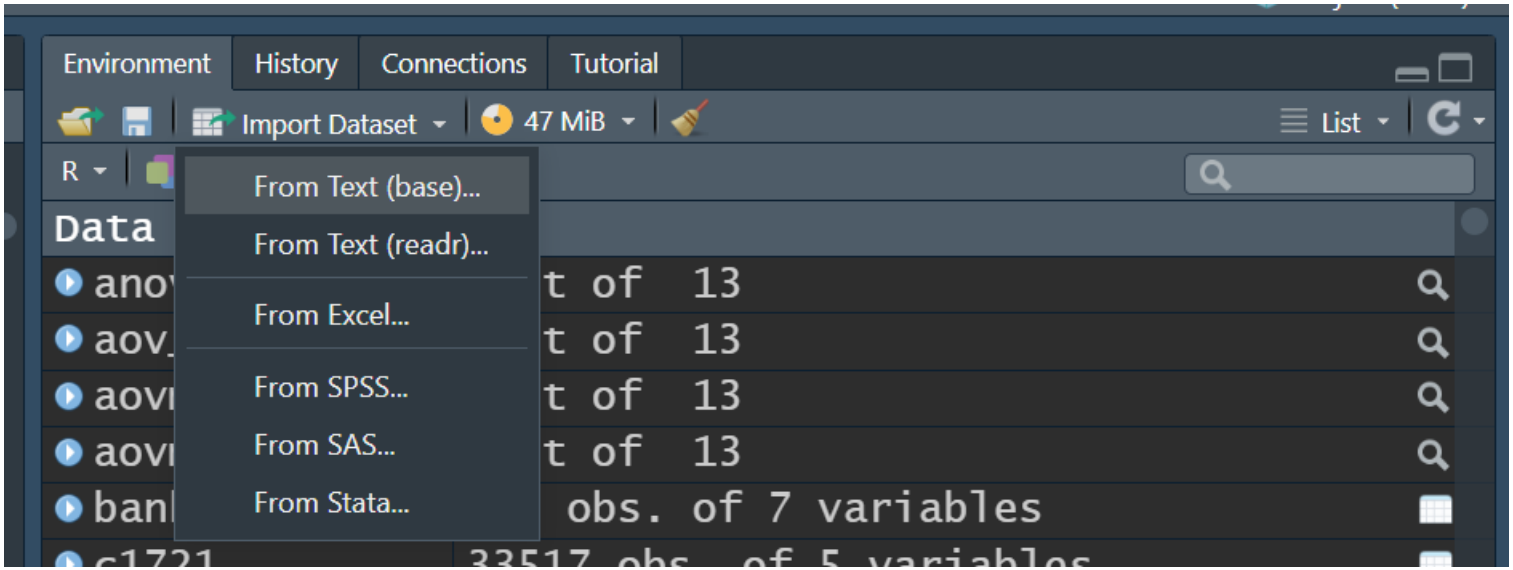
3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

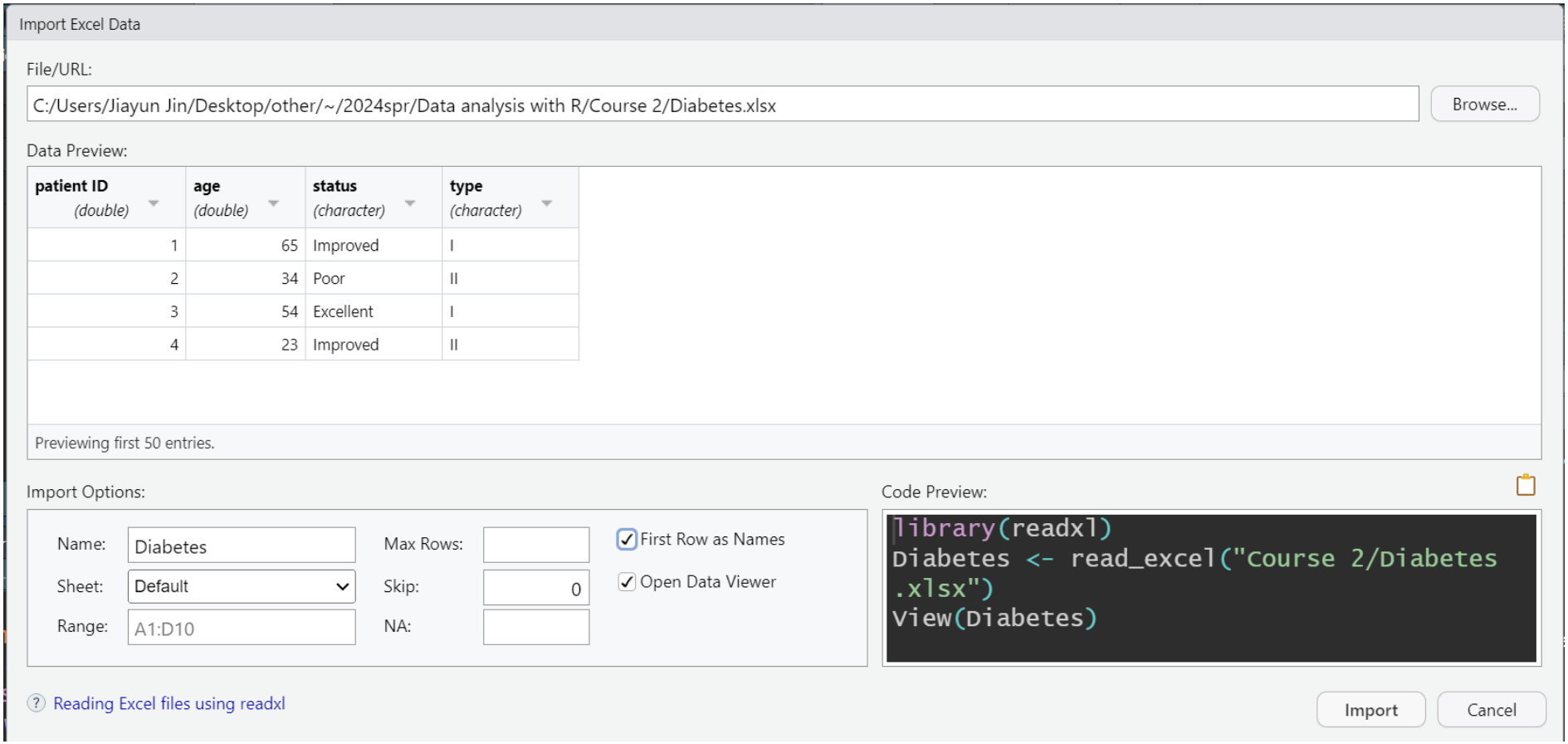
6 Summary

Data input


Environment
History
Connections
Tutorial
Import Dataset
47 MiB
List
R
From Text (base)...
From Text (readr)...
From Excel...
From SPSS...
From SAS...
From Stata...
Data
t of 13
t of 13
t of 13
t of 13
obs. of 7 variables

Data input

- Import Excel data



1 Understanding datasets

2 Data structures

3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

6 Summary

Annotating datasets

1. Variable labels

```
colnames(df)[i] <- "NewLabel"
```

```
> colnames(df_patient)
[1] "PatientID" "age"       "status"
> colnames(df_patient)[1]
[1] "PatientID"
> colnames(df_patient)[1] <- "ID"
> colnames(df_patient)
[1] "ID"     "age"    "status"
>
```

Annotating datasets

2. Value labels

Variable type

- Categorical variable
  - Nominal(名义变量) denotes labels or categories:
    E.g., gender (female, male), blood (A, B, AB, O)
  - Ordinal(有序变量): values have a natural order or ranking.
    E.g, satisfaction (very unsatisfied, unsatisfied, neutral, satisfied, and very satisfied)
- Numerical variable
  - Discrete(离散变量): only take integer values.
    E.g., number of children in a family, number of books sold
  - Continuous(连续变量): can take any value within a given range
    E.g., temperature, age

Annotating datasets

2. Value labels

ID	age	status	gender	type
1	23	Improved	1	I
2	34	Poor	2	II
3	53	Excellent	2	I
4	65	Improved	1	II

Categorical variables are called factors in R

- Categorical variable
  - Nominal: Type, gender
  - Ordinal: Status
- Numerical variable
  - Continuous: Age

Factors

Create value labels for a factor:

```
df_patient$gender <- factor(df_patient$gender,
                            levels = c(1, 2),
                            labels = c("male", "female"))
```

- levels indicate the actual values of the variable
- labels refer to the desired labels (meanings).

ID	age	status	gender	type
1	23	Improved	male	I
2	34	Poor	female	II
3	53	Excellent	female	I
4	65	Improved	male	II

1 Understanding datasets

2 Data structures

3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

6 Summary

Useful functions for working with data objects

Function	Purpose
length(object)	Gives the number of elements/components.
dim(object)	Gives the dimensions of an object.
str(object)	Gives the structure of an object.
class(object)	Gives the class of an object.
cbind(object1, object2)	Combines objects by column.
rbind(object1, object2)	Combines objects by row.
head(object)	Lists the first part of an object.
tail(object)	Lists the last part of an object.

1 Understanding datasets

2 Data structures

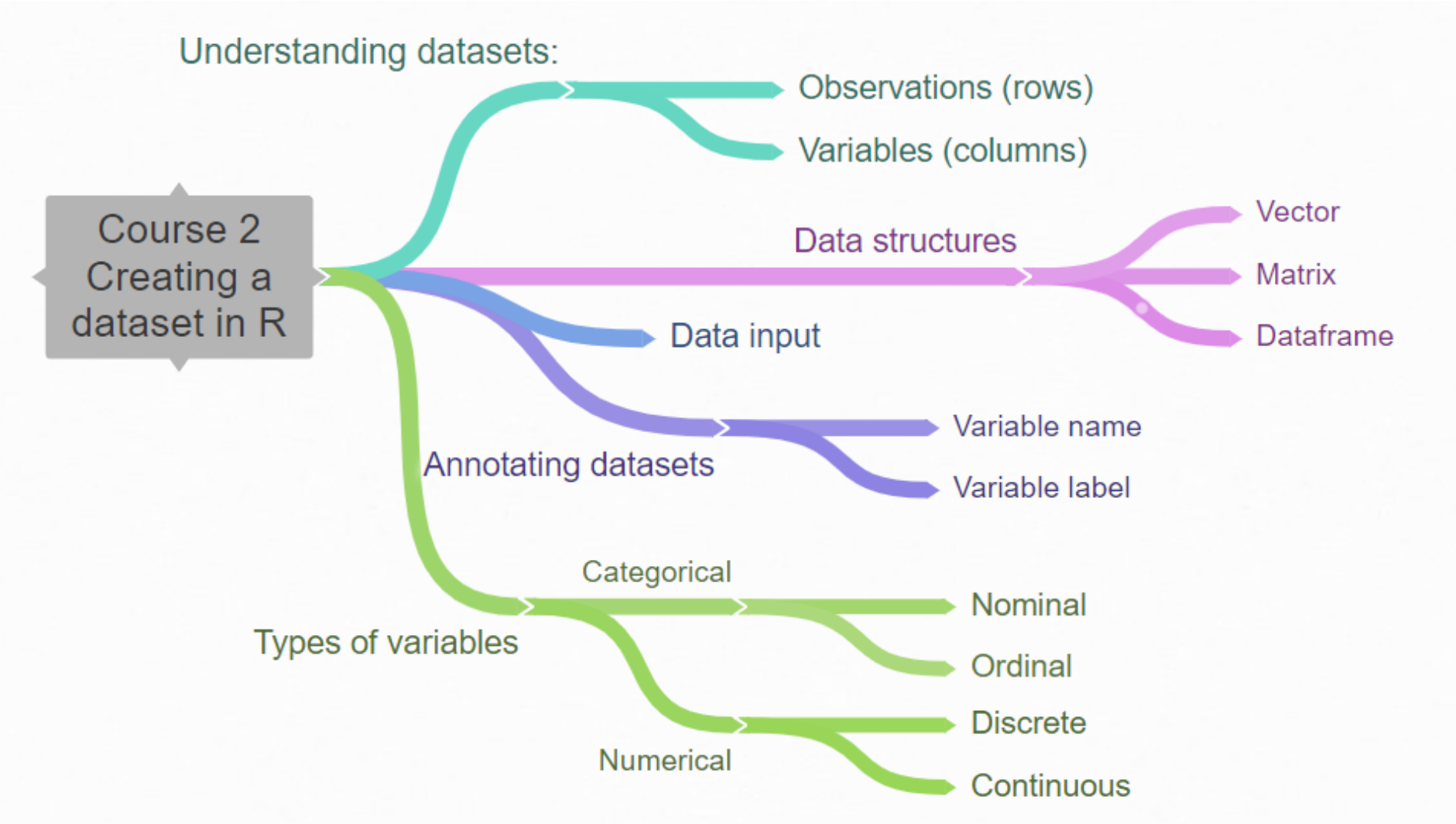
3 Data input

4 Annotating datasets

5 Useful functions for working with data objects

6 Summary

Summary


Course 2
Creating a
dataset in R
Understanding datasets:
Observations (rows)
Variables (columns)
Data structures
Vector
Matrix
Dataframe
Data input
Annotating datasets
Variable name
Variable label
Types of variables
Categorical
Nominal
Ordinal
Numerical
Discrete
Continuous

Exercise

1. Import data from 'Diabetes.xlsx' (Remember to install the relevant package first)
2. change it to a data frame ('as.data.frame')
3. Walk through the codes on Page 16 - 18.
4. Try the codes on Page 26.
5. Play with the functions on Page 28.

Have fun coding!